

DIKON JEROME KELAART

Mechanical Engineering Undergraduate, University of Moratuwa

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PROFESSIONAL SUMMARY

Final-year Mechanical Engineering undergraduate at the University of Moratuwa with practical experience across robotics, automation, machine design and IoT. During industrial training at a vehicle assembly plant, delivered an Arduino-based cone pick-and-place rig and a real-time IoT bolt-inventory dashboard for the production line. Skilled in SolidWorks, ANSYS, COMSOL, MATLAB and embedded development, and seeking an entry-level role in robotics and automation, industrial, or design and manufacturing engineering.

EDUCATION

University of Moratuwa

B.Sc. Engineering (Hons.) in Mechanical Engineering

Moratuwa, Sri Lanka

2022 – 2026

- Relevant coursework: Machine Design, Mechanisms, Fluid Dynamics, Thermodynamics, Manufacturing, Automation, Robotics and 3D Modelling.

St. Peter's College, Colombo

GCE Advanced Level – Physical Science Stream (Maths, Physics, Chemistry)

2019

2A, 1B

ENGINEERING PROJECTS

Vibration Energy Harvester for a GPS Vehicle Tracker – Final-Year Capstone

Semester 8, 2026

Designed and built an electromagnetic vibration energy harvester to recharge a GPS-tracker battery from vehicle-suspension vibration: acquired road-vibration data (ADXL345/ESP32), identified the dominant excitation frequency through MATLAB FFT/PSD analysis, modelled the coupled mechanical-electromagnetic system, validated it in COMSOL, and fabricated a SolidWorks prototype.

Skills: *SolidWorks, COMSOL Multiphysics, MATLAB, Arduino/ESP32, electromagnetic design, prototyping.*

Gel-Pen Assembly Line Optimisation, Atlas Axillia – Industrial Engineering

Semester 8, 2026

Conducted time studies and line-balancing on a gel-pen assembly line, identifying the manual sticker-pasting and card-packing stations as the main bottlenecks; validated the production flow in Siemens Tecnomatix Plant Simulation and proposed automation options supported by a cost-benefit and payback analysis.

Skills: *time study, line balancing, plant simulation, process improvement, cost-benefit analysis.*

Locally Value-Added Automotive Parts – Industrial Project Management

Semester 7, 2025

Prepared a design-and-manufacturing feasibility study for locally producing automotive components (injection moulding and sheet-metal forming, DFM, ISO 9001/TS 16949), covering the technical, financial (NPV, IRR, payback) and scheduling (WBS, CPM) analysis.

Skills: *SolidWorks, DFM, manufacturing processes, project management (WBS/CPM), financial analysis.*

Automated Road-Cone Pick-and-Place Machine

Semester 6, 2025

Industrial Training – Abans Auto (Pvt) Ltd, Hyundai Assembly Plant

Designed the mechanical system and Arduino-based control electronics for a machine that automatically picks and places road cones from a moving lorry, using encoder distance tracking, stepper positioning, a servo gripper and push-button mode switching.

Skills: *Arduino, embedded C/C++, mechatronics, motion control, sensors and actuators.*

Real-Time Bolt-Counting Kanban Dashboard

Semester 6, 2025

Industrial Training – Abans Auto (Pvt) Ltd, Hyundai Assembly Plant

Developed a sensor-based bolt-counting system with a real-time web dashboard (ESP32, Node.js, Socket.io) displaying live per-bin stock by part number on the assembly line.

Skills: *ESP32, IoT, Node.js/Express, Socket.io, JavaScript, real-time dashboards.*

SME Operations & Sustainability Improvement Study – Social/Community Project

Semester 6, 2025

Analysed a bakery SME's operations and recommended process automation, energy-efficiency measures (heat-recovery ovens, rooftop solar PV, ISO 50001 energy management) and a POS/inventory system to support demand forecasting.

Skills: *operations analysis, process automation, energy efficiency, sustainability, demand forecasting.*

Coconut-Processing Production-Line Optimisation – Production & Operations Management

Semester 5, 2024

Modelled and optimised a coconut-processing line in Siemens Tecnomatix Plant Simulation using lean and just-in-time principles with demand forecasting, identifying opportunities to reduce cutting time, material wastage and labour requirements.

Skills: *Siemens Tecnomatix Plant Simulation, lean / JIT, process optimisation, demand forecasting.*

Sugar-Cane Grinding Machine Gearbox – Machine Design Project

Semester 5, 2024

Designed a multi-speed sliding-mesh gearbox (three forward, one reverse) for a sugar-cane mill, sizing and verifying the gears, shafts, bearings, keys and V-belt drive against static, dynamic and wear loading, with a complete SolidWorks model.

Skills: *SolidWorks, machine design, gear/shaft/bearing design, design under combined loading.*

Reverse Engineering of a Floodlight

Semester 4, 2024

Reverse-engineered a commercial floodlight, producing part models, assemblies and production drawings to study its design and manufacturing process.

Skills: *SolidWorks, reverse engineering, 3D modelling, production drawings.*

Smart Waste Management System

Semester 2, 2023

Built an automated waste classification, compression and sealing system using Arduino-based machine learning and image processing, with a companion IoT monitoring app.

Skills: *Arduino, Python, machine learning, image processing, IoT.*

TECHNICAL SKILLS

CAD & Design: SolidWorks, Solid Edge, parametric modelling, production drawings, reverse engineering

Simulation & Analysis: ANSYS (FEA), COMSOL Multiphysics, MATLAB/Simulink, Siemens Tecnomatix Plant Simulation

Automation & Control: PLC, pneumatics, control systems, P&ID, Arduino, ESP32, stepper/servo motors, encoders, IoT

Robotics & Mechatronics: pick-and-place systems, sensors, actuators, motion control

Programming: Python, C++, JavaScript (HTML/CSS), MATLAB, Node.js/Express, React

Languages: English (fluent), Sinhala (native)

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Team FalconE Racing – Formula Student

Steering Engineer

Dec 2025 – Apr 2026

Steering Team Member

May 2024 – Nov 2025

University of Moratuwa Mechanical Engineering Society (UMMES)

Vice-President

Nov 2024 – Present

Department of Mechanical Engineering

Batch Representative – Semesters 4 & 5

IEEE Student Branch, University of Moratuwa

Finance Committee Member

Nov 2023 – Present

EXMO Flagship Engineering Exhibition

Logistics Team Member

2023

REFERENCES

Dr. M. Hordagoda

BSc Eng (Moratuwa), PhD (Tampa, USA)

Applied Mechanics

QA Coordinator, Department of Mechanical Engineering

Faculty of Engineering, University of Moratuwa

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